

CS 280 Paper Presentation

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Outline

- 1 Financial time series forecasting using support vector machines
- 2 Credit rating analysis with support vector machines and neural networks: a market comparative study

Financial time series forecasting using support vector machines

Kyoung-jae Kim, Dongguk University. 2013

- This study uses SVM to predict the stock price index
- Compared performance of SVM, ANN and case-based reasoning for stock market prediction

Introduction

- ANN has inconsistent performance when faced with noisy data and complex dimensionality
- Backprop algorithm still needs to determine several parameters such as input features, hidden layer size, learning rate and momentum term
- ANN implements the *empirical risk minimization principle* while SVM implements the *structural risk minimization principle*

Methodology

- Data used is technical indicators and direction of change in daily stock price index: 80% for training, 20% for validation
- There are 12 attributes that determine the direction of daily change, scaled into $[-1.0, 1.0]$ and classified to '0' or '1'
- Polynomial and RBF kernels are used for SVM; three-layer network for ANN and nearest-neighbor method for CBR to retrieve relevant cases

Methodology

Why are the input values scaled into $[-1, 1]$?

The goal of linear scaling is to independently normalize each feature component to ensure that larger values do not overwhelm smaller inputs to reduce prediction errors.

Methodology

Table 1
Initially selected features and their formulas

Feature name	Description	Formula	Refs.
%K	Stochastic %K. It compares where a security's price closed relative to its price range over a given time period.	$\frac{C_t - LL_{t-n}}{HH_{t-n} - LL_{t-n}} \times 100$, where LL_t and HH_t mean lowest low and highest high in the last t days, respectively.	[1]
%D	Stochastic %D. Moving average of %K.	$\frac{\sum_{i=0}^{n-1} \%K_{t-i}}{n}$	[1]
Slow %D	Stochastic slow %D. Moving average of %D.	$\frac{\sum_{i=0}^{n-1} \%D_{t-i}}{n}$	[9]
Momentum	It measures the amount that a security's price has changed over a given time span.	$C_t - C_{t-n}$	[3]
ROC	Price rate-of-change. It displays the difference between the current price and the price n days ago.	$\frac{C_t}{C_{t-n}} \times 100$	[16]
Williams' %R	Larry William's %R. It is a momentum indicator that measures overbought/oversold levels.	$\frac{H_t - C_t}{H_t - L_t} \times 100$	[1]
A/D Oscillator	Accumulation/distribution oscillator. It is a momentum indicator that associates changes in price.	$\frac{H_t - C_{t-1}}{H_t - L_t}$	[3]
Disparity5	5-day disparity. It means the distance of current price and the moving average of 5 days.	$\frac{C_t}{MA_5} \times 100$	[5]
Disparity10	10-day disparity.	$\frac{C_t}{MA_{10}} \times 100$	[5]
OSCP	Price oscillator. It displays the difference between two moving averages of a security's price.	$\frac{MA_5 - MA_{10}}{MA_5}$	[1]
CCI	Commodity channel index. It measures the variation of a security's price from its statistical mean.	$\frac{(M_t - SM_t)}{(0.015 D_t)}$ where $M_t = (H_t + L_t + C_t)/3$, $SM_t = \frac{\sum_{i=1}^n M_{t-i+1}}{n}$, and $D_t = \frac{\sum_{i=1}^n M_{t-i+1} - SM_t }{n}$	[13]
RSI	Relative strength index. It is a price following an oscillator that ranges from 0 to 100.	$100 - \frac{100}{1 + \frac{\sum_{i=0}^{n-1} Up_{t-i}(n)}{\sum_{i=0}^{n-1} Dn_{t-i}(n)}}$ where Up_t means upward-price-change and Dn_t means downward-price-change at time t .	[1]

C_t is the closing price at time t , L_t the low price at time t , H_t the high price at time t , MA_t the moving average of t days.

Results

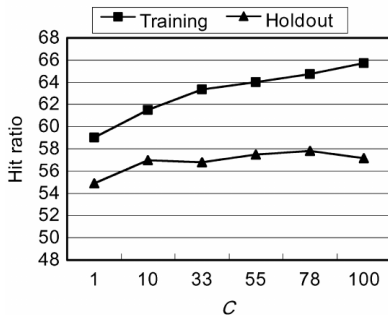


Fig. 1. The results of SVMs with various C where δ^2 is fixed at 25.

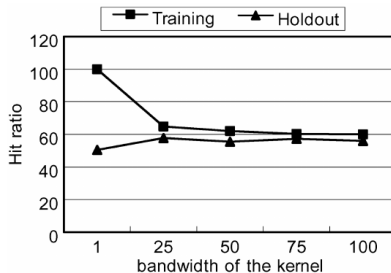


Fig. 2. The results of SVMs with various δ^2 where C is fixed at 78.

Results

Table 5

The best prediction performances of SVM, BP, and CBR (hit ratio: %)

	SVM	BP	CBR
Training data	64.7526	58.5217	
Holdout data	57.8313	54.7332	51.9793

- SVM leads to better generalization than conventional techniques when used with optimal parameters

Credit rating analysis with support vector machines and neural networks: a market comparative study

*Z. Huang, H. Chen, C. Hsu, W. Chen, S. Wu,
University of Arizona. 2003*

- Used SVM, ANN and CBR to predict corporate credit ratings based on input financial variables
- Obtained relative importance of input variables from the neural network models
- Conducted a market comparative analysis on the determining factors in US and Taiwan markets

Introduction

- Credit ratings is a measure of riskiness of bonds: lower rating indicates higher risk
- Typically very costly to obtain, needs large amount of time and human resources to perform deep analysis
- Rating agencies emphasize the importance of analysts' highly subjective judgment in determining credit ratings

Introduction

Why are intelligent models important to credit rating analysis?

It help users capture fundamental characteristics of different financial markets and which attributes have more contribution than others. In other terms, to reverse engineer the subjective judgment done by credit raters.

Methodology

Methodology

Table 2
Ratings in the two data sets

TW data		US data	
twAAA	8	AA	20
twAA	11	A	181
twA	31	BBB	56
twBBB	23	BB	7
twBB	1	B	1
Total	74	Total	265

Methodology

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Table 3
Financial ratios used in the data set

	Financial ratio name/description	ANOVA between-group <i>p</i> -value
X1	Total assets	0.00
X2	Total liabilities	0.00
X3	Long-term debts/total invested capital	0.12
X4	Debt ratio	0.00
X5	Current ratio	0.36
X6	Times interest earned (EBIT/interest)	0.00
X7	Operating profit margin	0.00
X8	(Shareholders' equity + long-term debt)/fixed assets	0.00
X9	Quick ratio	0.37
X10	Return on total assets	0.01
X11	Return on equity	0.04
X12	Operating income/received capitals	0.00
X13	Net income before tax/received capitals	0.00
X14	Net profit margin	0.00
X15	Earnings per share	0.00
X16	Gross profit margin	0.02
X17	Non-operating income/sales	0.81
X18	Net income before tax/sales	0.00
X19	Cash flow from operating activities/current liabilities	0.84
X20	(Cash flow from operating activities/capital expenditures + increased in inventory + cash dividends) in last 5 years	0.64
X21	(Cash flow from operating activities-cash dividends)/ (fixed assets + other assets + working capitals)	0.08

Results

Table 4

Prediction accuracies (LogR: logistic regression model, SVM: support vector machines, NN: neural networks)

	10-fold Cross-validation			Leave-one-out Cross-validation		
	LogR (%)	SVM (%)	NN (%)	LogR (%)	SVM (%)	NN (%)
TW I	72.97	79.73	75.68	75.68	79.73	74.32
TW II	70.27	77.03	75.68	70.27	75.68	74.32
US I	76.98	78.87	80.00	75.09	80.38	80.75
US II	75.47	80.00	79.25	75.47	80.00	75.68

- Relatively small list of financial variables largely determined the rating results

Results

Results

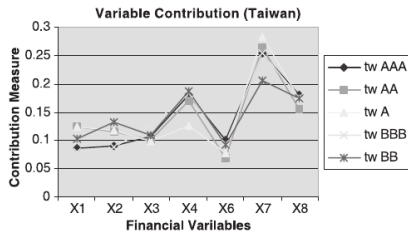
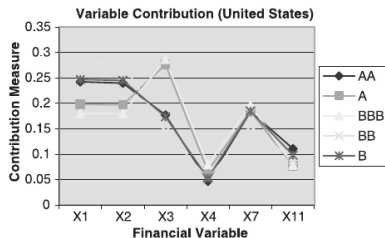
$$\text{Con}_{ik} = \frac{\sum_{j=1}^J \frac{|w_{ji}| |v_{jk}|}{\sum_{i=1}^I |w_{ji}|}}{\sum_{i=1}^I \sum_{j=1}^J \frac{|w_{ji}| |v_{jk}|}{\sum_{i=1}^I |w_{ji}|}}$$

- Garson's measure of relative contribution of input i to output k
- Assessed relative importance of each input variables to the bond-rating process in different markets

Results

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Thank you for your attention!